



Scuola Superiore
Sant'Anna

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Bioinspired and Soft Robotics

Soft Robotics Technologies

Electro- Magneto-Rheological Fluids

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History

Although the practical study and application of field-responsive fluids is relatively new (the last two decades), their electromagnetic properties have been known since the late-1940s.

Electrorheology and magnetorheology, in particular, were discovered by Winslow (1949) and Rabinow (1948) almost simultaneously.



Definition

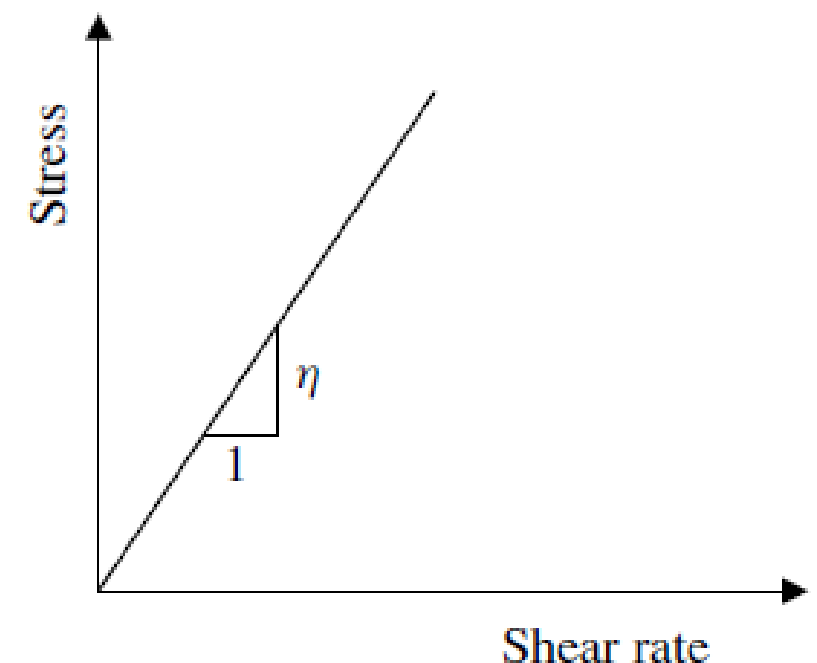
The term rheology defines the science of the deformation and flow of matter

Rheology is the science that studies the elastic, plastic and viscous properties of materials. Rheology is particularly interested in describing the stress–strain relationship in flowing fluids. The mathematical expressions formulating the stress–strain relationship in fluids are known as *rheological equations of state*.

Fluids are divided into two broad categories: **Newtonian** and **non-Newtonian**.

1. *Newtonian fluids*. In Newtonian fluids, rheological equations of state describe a linear relationship between shear stress and rate of shear as in the following expression:

$$\tau = \eta \dot{\gamma}$$



Definition

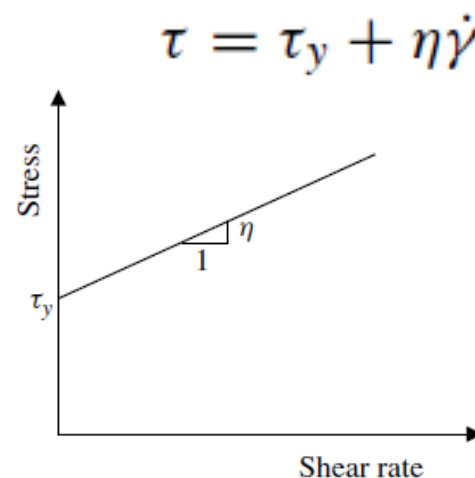
2. *Non-Newtonian fluids*. Within this category, all the remaining fluids are classified. Roughly, they can be, in turn, included in one of the following groups:

ERF - MRF → (a) *Time-independent fluids*. Here, the shear stress is only a function of shear rate, τ is a function of $\dot{\gamma}$.

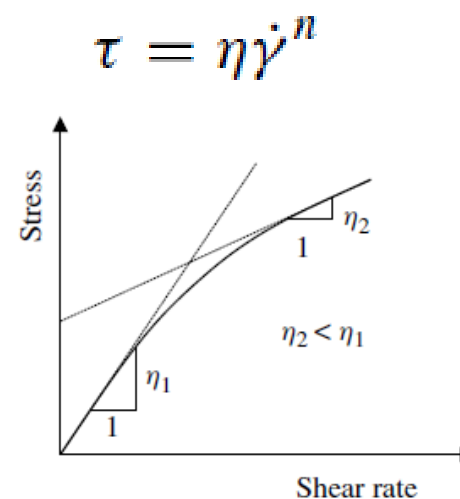
(b) *Time-dependent fluids*. Rheological equations of state depend on how the fluid has been sheared and on previous history.

(c) *Elastoviscous fluids*. These fluids are predominantly viscous but exhibit partial elastic recovery after deformation.

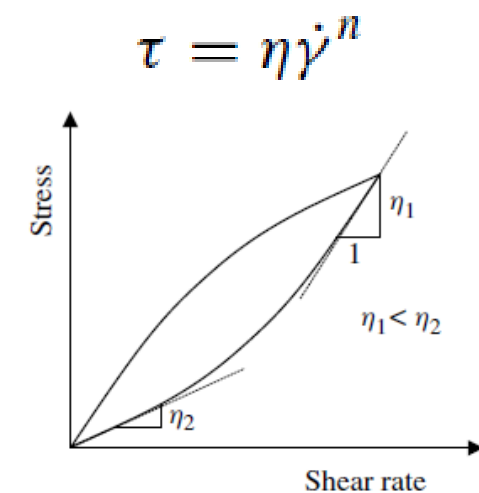
Bingham Fluids. A Bingham fluid is an idealized material whose internal structure collapses above a yield stress τ_y . For higher stresses, the rheological behavior is linear.

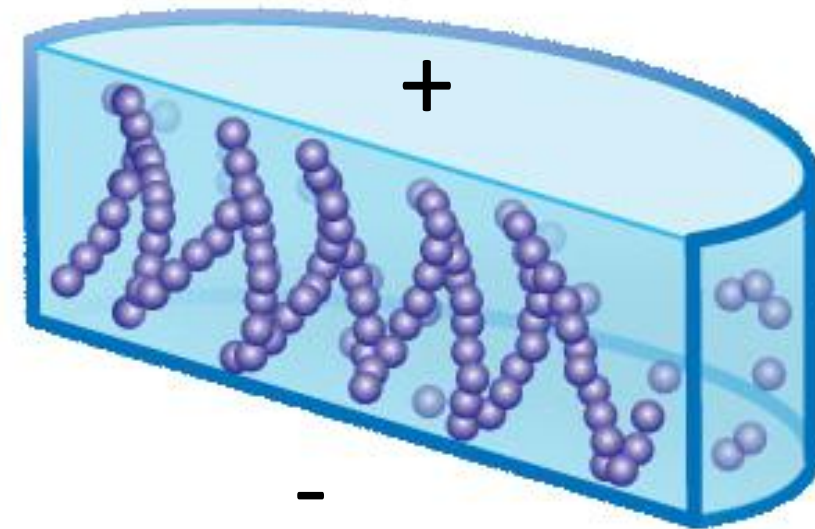
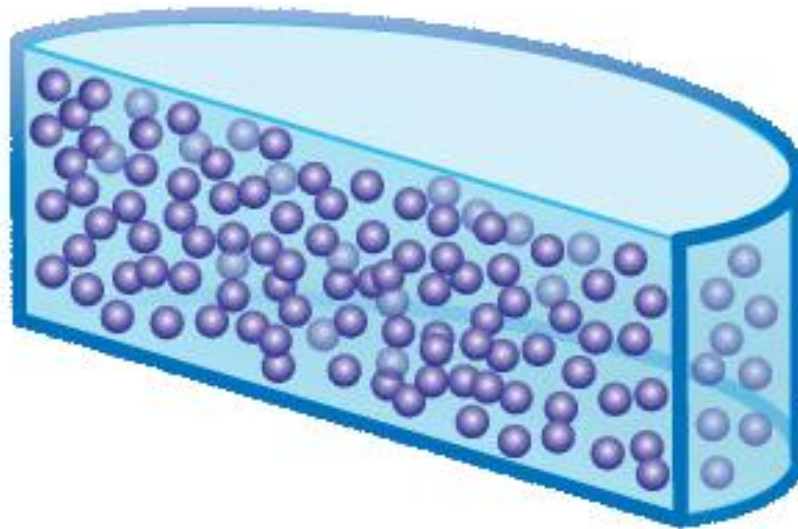


Pseudoplastics. These fluids exhibit no yield stress, $\tau_y = 0$, but the stress increases less than linearly with the rate of strain. ($n < 1$)



Dilatant fluids. The curve for a dilatant fluid looks the same as for a pseudoplastic fluid, except that the slope of the stress–rate of strain relationship increases. ($n > 1$)



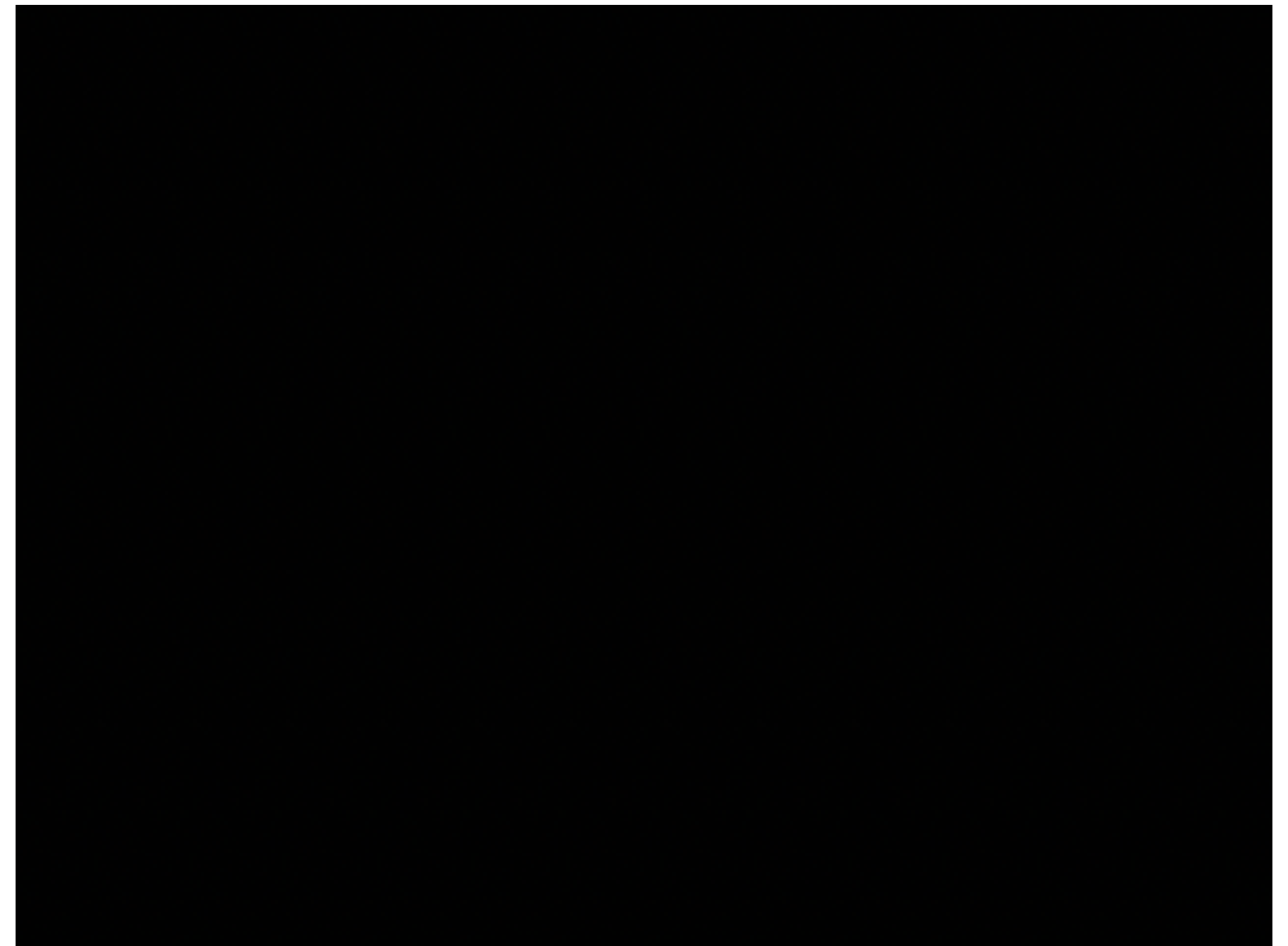


Ferrofluids

Ferrofluids are colloidal suspensions of single-domain magnetic particles in either aqueous or nonaqueous liquids. The typical particle size in ferrofluids is of the order of a **few** (commonly 5 to 10) **nanometers**.

Ferrofluids undergo field-induced viscosity changes but **do not present yield stress**. This is basically because the magnetic particles are ultra small: thermal agitation of the particles produces Brownian forces that prevent the formation of interparticle chains.

The application of an external magnetic field generates body forces in the ferrofluid, which in turn cause changes in the viscosity (which can double because of this effect). Such fluids are **unsuitable** for use as actuator technologies owing to the high cost of achieving only slight modification of rheological properties.



Magnetorheological fluids

Magnetorheological fluids are suspensions of noncolloidal, multidomain and magnetically soft particles in base organic or aqueous liquids. The size of the magnetic particles suspended in the base liquid may vary from **0.1 to 10 μm** .

Magnetorheological fluids behave similarly to Bingham plastics. When not subjected to any external magnetic field, the apparent viscosity of MR fluids is of the order of 0.1 to 10 Pa·s. Where an external magnetic field is applied, within a very short response time, of the order of milliseconds, the apparent viscosity will **undergo an abrupt change of several (up to 5–6) orders of magnitude**.

The fraction concentration of magnetic particles in the base liquid can vary amongst different MR fluids. Typical values for the fraction concentration are of the order of 20 to 60%. It is believed that the increased yield stress and apparent viscosity is due to the build up of **magnetic particle chains in response to the interparticle magnetic interaction**.

Where a low shear stress is applied to the MR fluid, particle chains will first be stretched and deformed. If the applied stress is increased further, chains will eventually break and the material will start to flow.

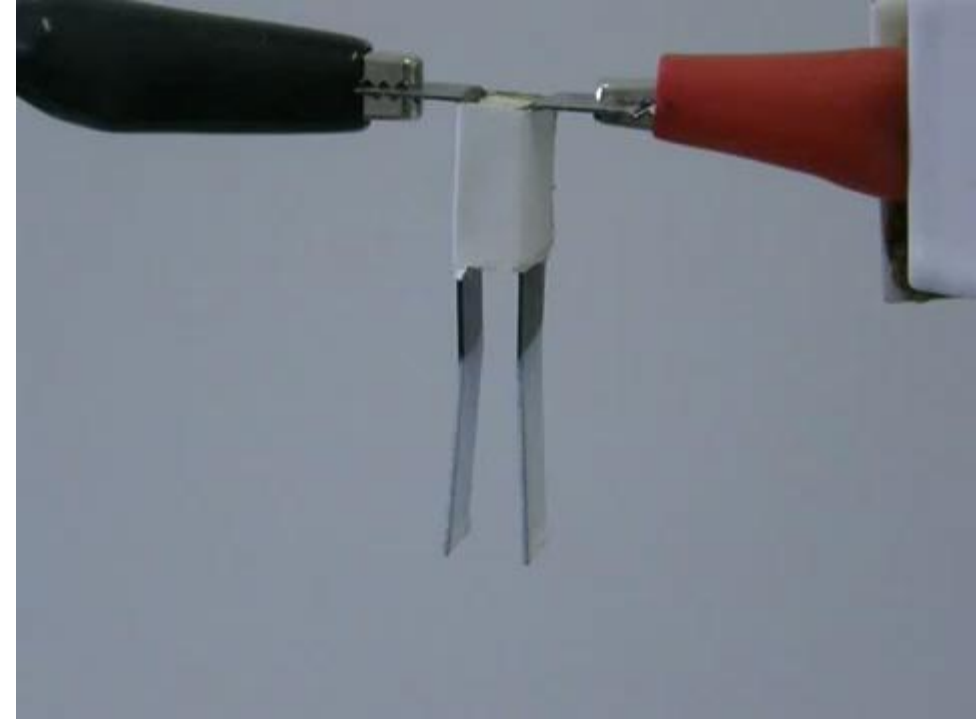
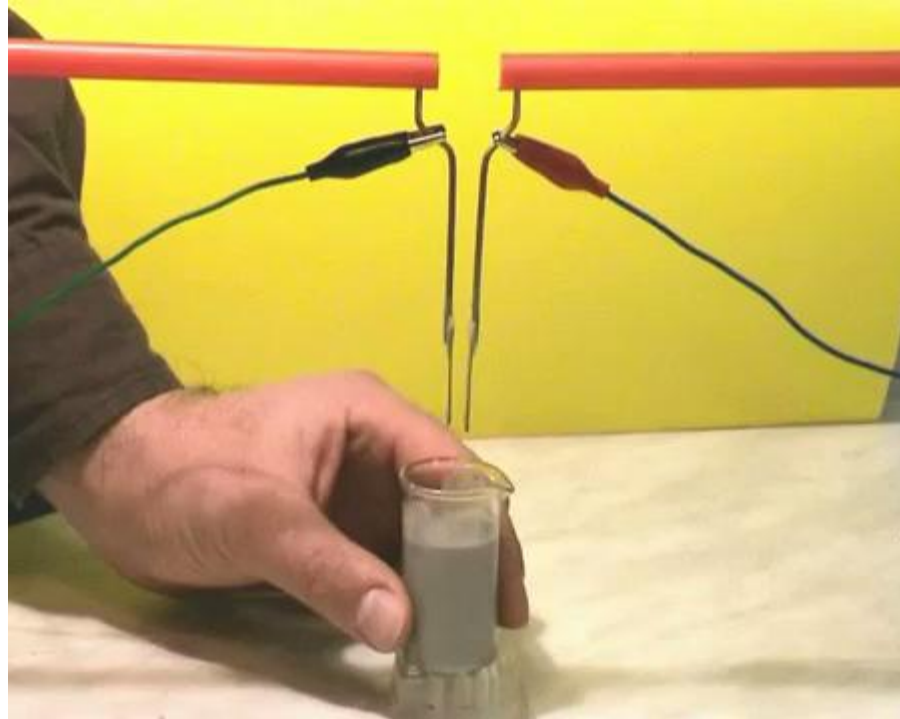


Electrorheological fluids

Electrorheological (ER) fluids are suspensions of electrically active particles in an electrically insulating liquid. Particle sizes in ER fluids are, on average, **one order of magnitude larger than particle sizes in MR fluids** (between 1–100 μm).

Similar to MR fluids, the electrically active particles in ER fluids are subject to interparticle interactions causing the formation of particle chains. Again, ER fluids conform to a Bingham model, but **the maximum yield stress is 2 orders of magnitude lower in ER fluids than in MR fluids**.

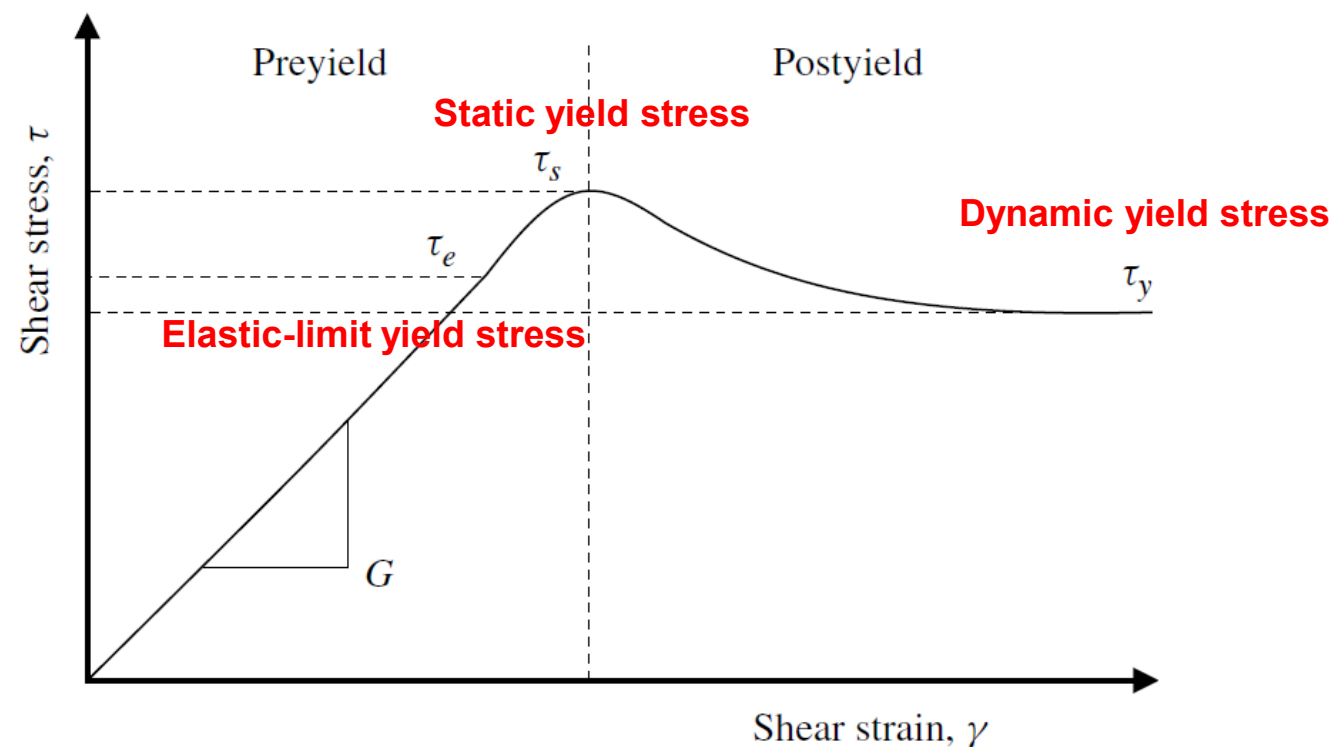
Maximum yield stress values are typically of the order of 1 to 5 kPa. The maximum applicable electric field in ER fluids is limited by the breakdown of the dielectric characteristics of the fluid.



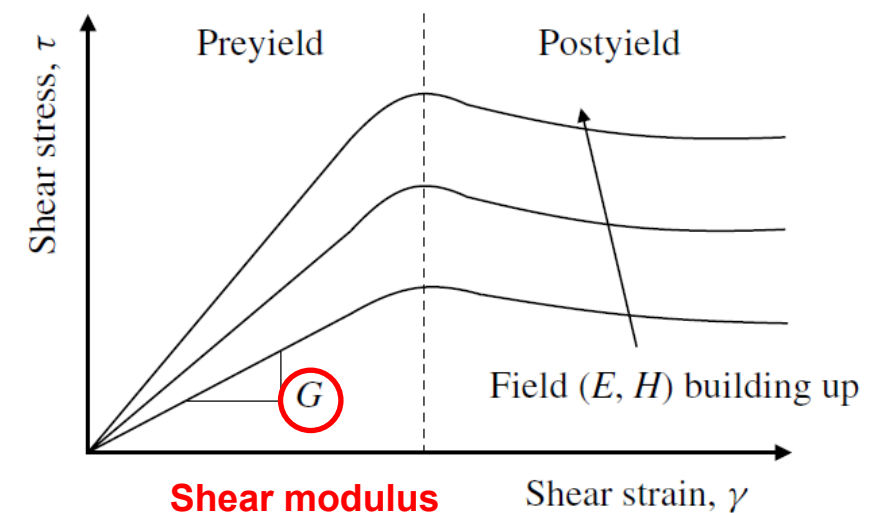
Electro- Magneto-rheological fluids

The electrorheological effect (ERE) and the magnetorheological effect (MRE) can be defined respectively as the increase in shear stress because of an applied electric and magnetic field.

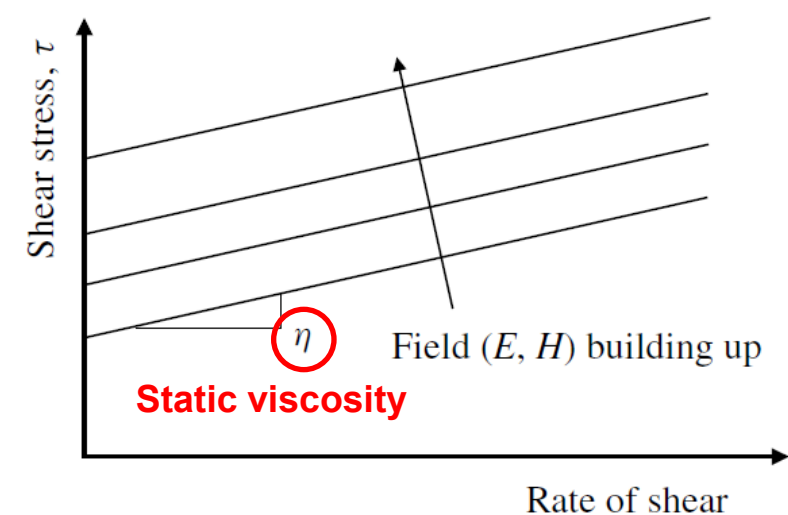
Mechanical operation with ER or MR fluids can be analyzed with respect to two situations: preyield and postyield operation. Both situations depend on the level of mechanical stress being exerted on the fluid.



$$\tau = G\gamma, \dot{\gamma} = 0 \quad \tau < \tau_e \quad \text{the fluid behaves as a solid}$$



$$\tau = \tau_y + \eta\dot{\gamma} \quad \tau \geq \tau_s \quad \text{the material flows}$$



Mechatronic design concepts

The ratio of the field-dependent term at the maximum admissible field, $X_\tau(H_{\max})$ or $X_\tau(E_{\max})$, to the viscosity term, $X_\eta(Y)$, is known as the **mechanical control ratio**, λ :

$$\lambda_E = \frac{X_\tau(E_{\max})}{X_\eta(Y)}$$

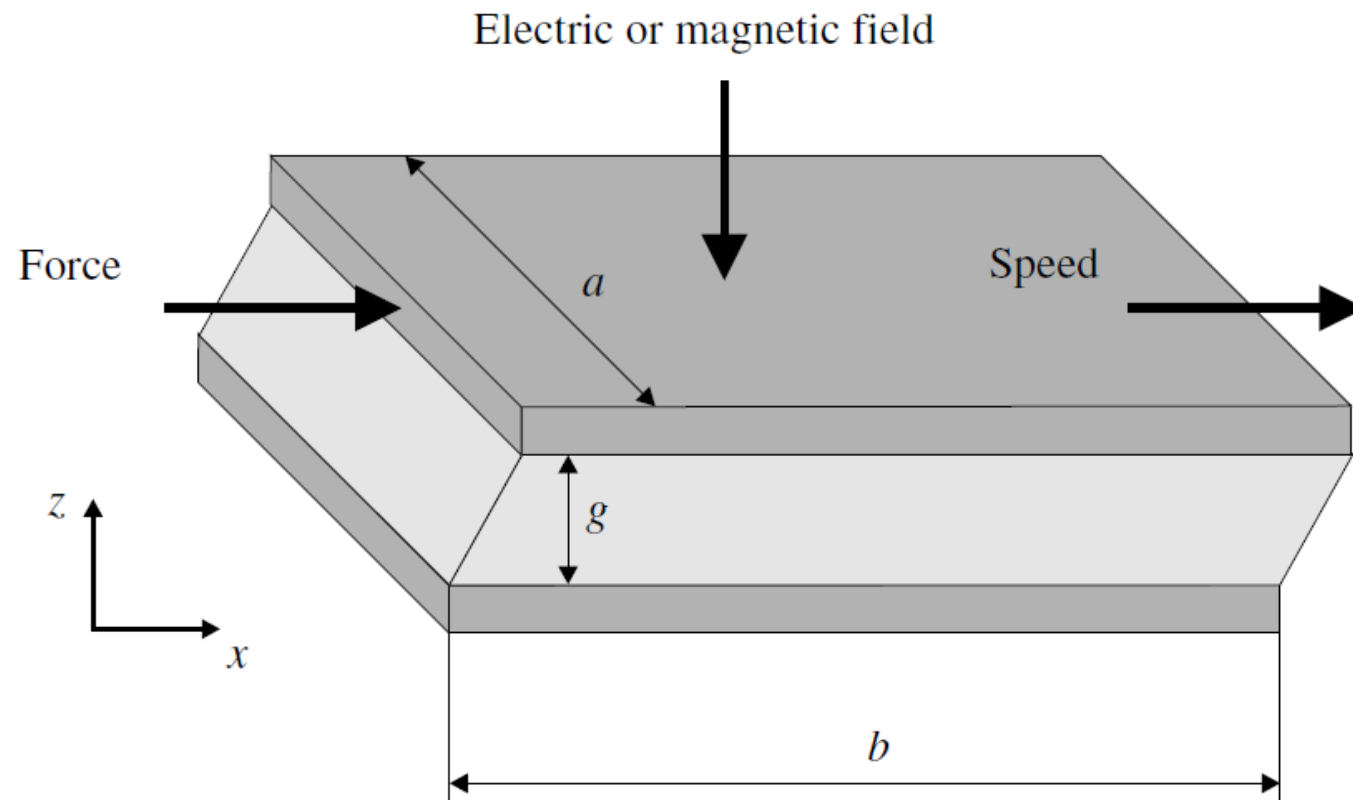
$$\lambda_H = \frac{X_\tau(H_{\max})}{X_\eta(Y)}$$

The mechanical control ratio is an indication of the ratio of the dissipative force in the 'ON' state to the dissipative force in the 'OFF' state. The design objective is typically to achieve the maximum dissipation in the 'ON' state while allowing minimum dissipation when there is no external field.

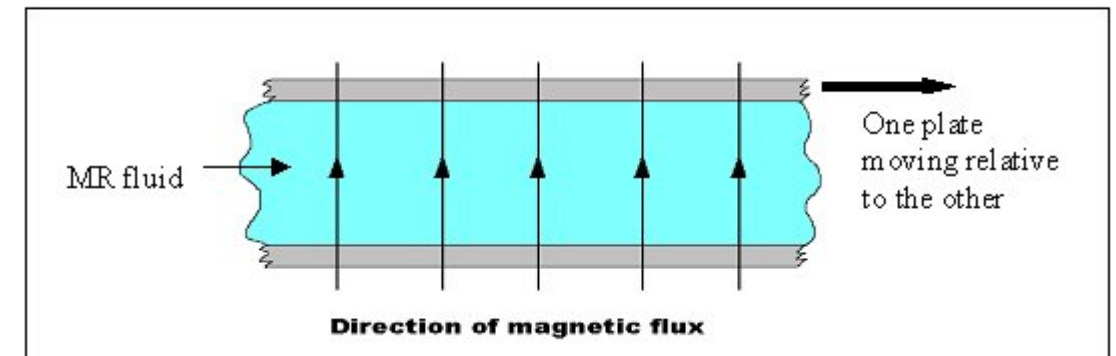


Mechatronic design concepts

Shear (or clutch) mode device



Shear Mode



$$F_{\eta} = \frac{\eta \dot{x} b a}{g}$$

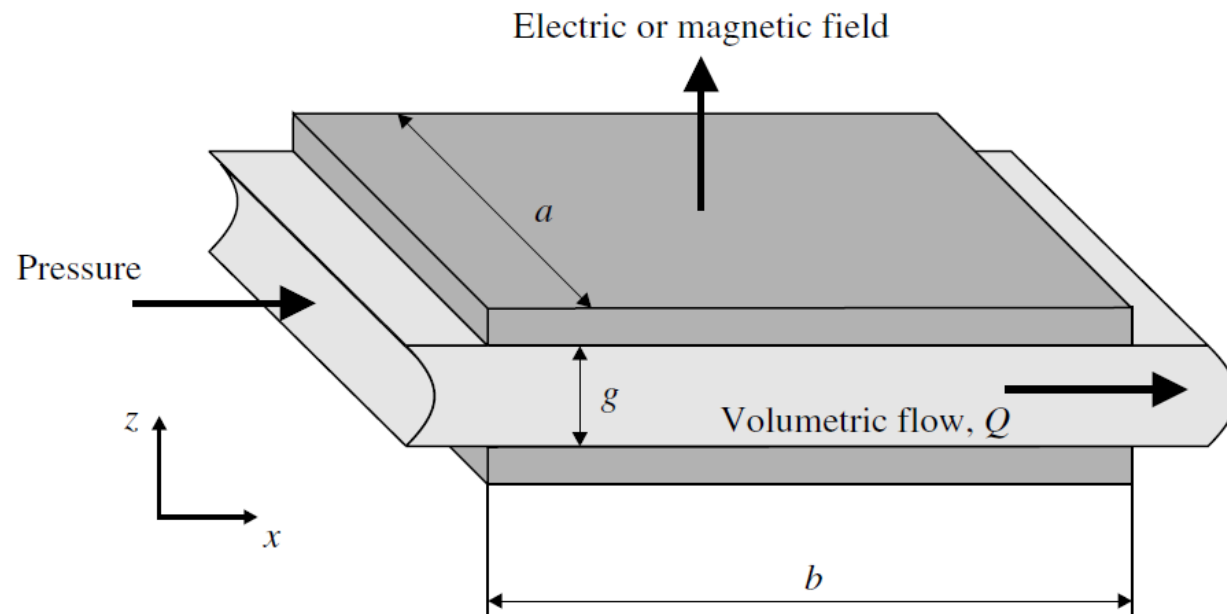
$$F_{\tau}(E, H) = \tau_y(E, H) b a$$

One of the plates is considered fixed, the other is subject to a relative displacement, and the distance between them is constant.

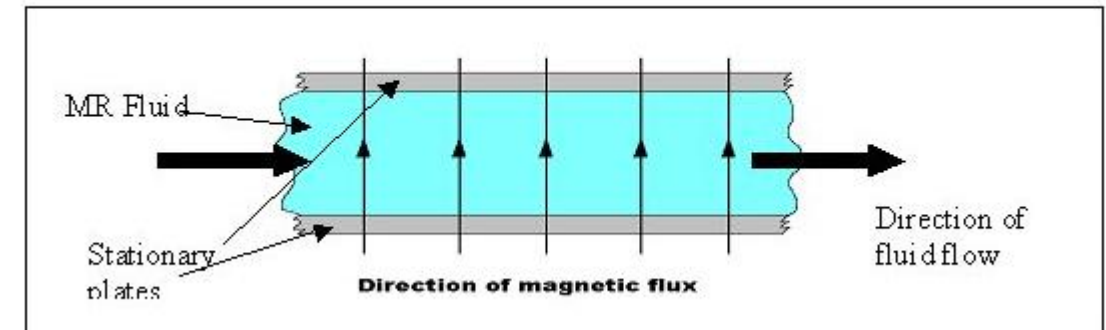


Mechatronic design concepts

Flow (or valve) mode device



Flow mode



Both plates are fixed in this configuration, so that a pressure drop, $X(E) = P(E)$, forces the fluid to flow at a volumetric flow rate, $Y = Q$.

$$P_{\eta}(S) = \frac{12\eta Qb}{g^3 a}$$

$$P_{\tau}(E, H) = \frac{c\tau_y(E, H)b}{g}$$

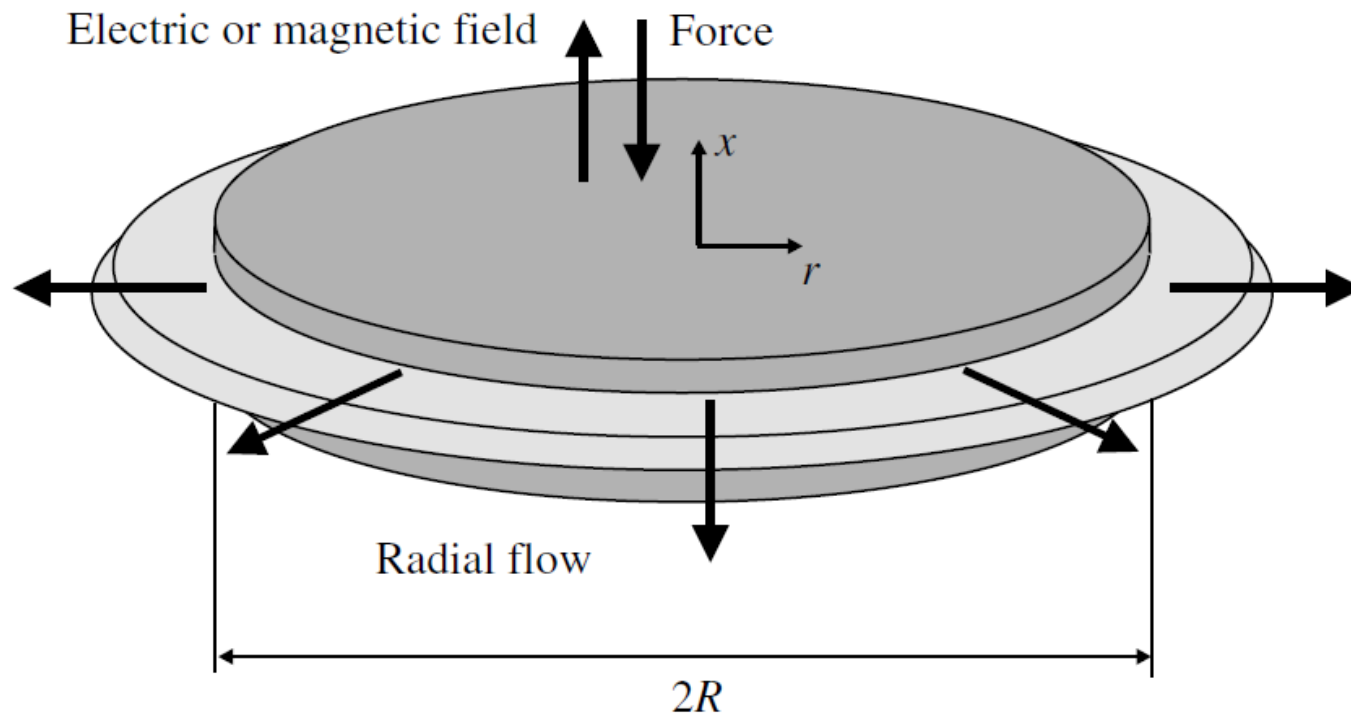
$$c = 2 \quad \text{for} \quad \frac{\Delta p_{\tau}}{\Delta p_{\eta}} \leq 1,$$

$$c = 3 \quad \text{for} \quad \frac{\Delta p_{\tau}}{\Delta p_{\eta}} > 100.$$

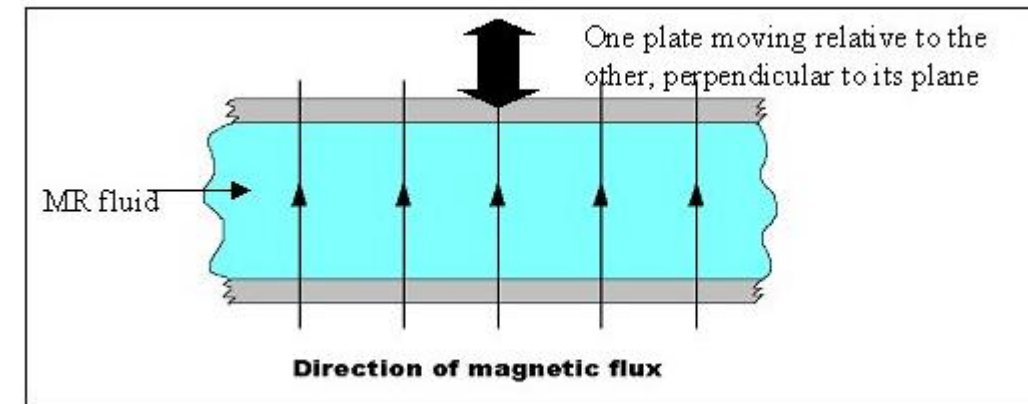


Mechatronic design concepts

Squeeze mode design



Squeeze-Flow Mode



The design concept for squeeze mode devices is the only one that exploits the relative perpendicular displacement of the plates.

A force, $F(E)$, is applied perpendicularly to the plates. The fluid confined between the two plates is forced to flow horizontally out of the interplate gap.

$$F_{\eta} = \frac{3\pi \eta R^4 \dot{x}}{2(g_0 - x)^3}$$

$$F_{\tau}(E, H) = \frac{4\pi \tau_y(E, H) R^3}{3(g_0 - x)}$$



How to choose

Property	Units	MR Fluids	ER Fluids	Ferrofluids
<i>Physical properties and constituents</i>				
Particle type	–	Fe, ferrites	Zeolites, polymers, $BaTiO_3$	Ceramics, Fe, ferrites
Particle size	μm	0.05–10	0.5–100	$2–10 \cdot 10^{-3}$
Base liquid	–	nonpolar oils, polar liquids	Oils	Oils, water
Density	kg/m^3	$3–5 \cdot 10^3$	$1–2 \cdot 10^3$	$1–2 \cdot 10^3$
Max. yield stress	kPa	50–100	2–5	–
Operating temp.	C	–40 to +150	–25 to +125	–
No field viscosity	$\text{mPa}\cdot\text{s}$	100–1000	100–1000	2–500
<i>Electromagnetic properties</i>				
Maximum field	–	$\approx 250 \text{ kA/m}$	$\approx 4 \text{ kV/mm}$	–
Field-induced change	kPa	$\tau_y(H) \approx 10^2$	$\tau_y(H) \approx 5$	$\frac{\eta(H)}{\eta(0)} \approx 2$
Max. energy density	kJ/m^3	100	1	–
Response time	s	$< 10^{-3}$	$< 10^{-3}$	–



Electro- Magneto-rheological fluids



Electro- Magneto-rheological fluids

Applications

http://www.oandp.com/articles/2003-10_03.asp

